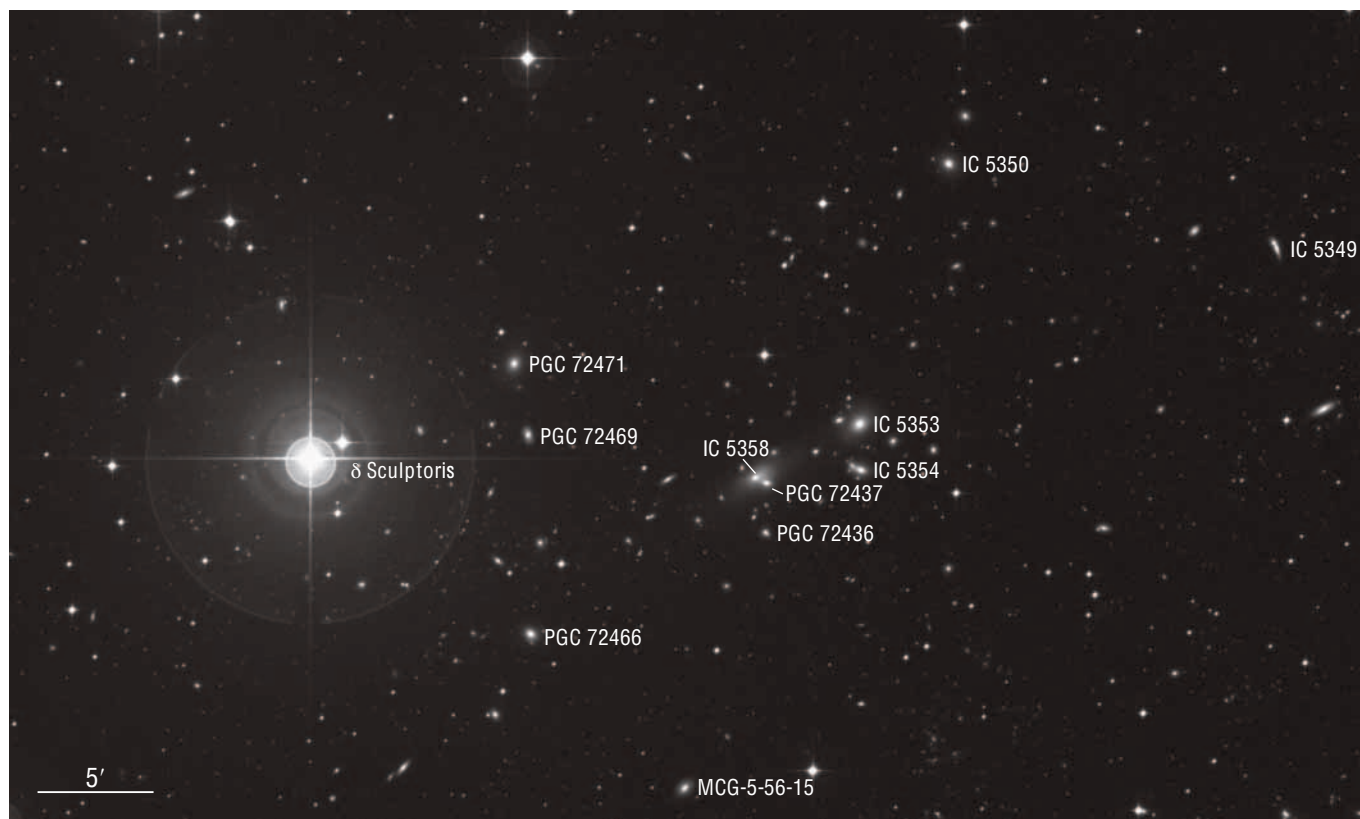




The rich galaxy cluster Abell 4038 awaits big scopes just west of the 4th-magnitude star Delta Sculptoris.

■ **deep-sky notebook** | By Steve Gottlieb

Exploring a Southern Galaxy Cluster

Galaxies are gregarious. On every scale they tend to huddle with neighbors, beginning with pairs and small groups and extending on up to rich clusters containing hundreds of members and spanning tens of millions of light-years. On even grander scales, sheets and filaments of clusters form *superclusters* up to several hundred million light-years long, separated by immense, bubble-like near-voids. Galaxy gatherings are the largest structures in the universe, and their origin, dynamics, and evolution form one of the hottest topics in cosmology.

How much of this action can amateurs see? Some groupings are obvious on sky atlases. Most famous is the Virgo Cluster, spanning the border of Virgo and Coma Berenices in the spring sky. It's the core of our Virgo Supercluster, which includes the Milky Way on its outskirts and shows up on Tirion and Sinnott's *Sky Atlas 2000.0* as a long band of red galaxy symbols spanning several charts from Ursa Major to Centaurus. Browse through the

deeper *Uranometria 2000.0* or *Millennium* sky atlases and you'll find many smaller groupings. Some of these make exciting targets for large amateur telescopes.

The first comprehensive survey of galaxy clusters was completed in 1958 by George Abell. He identified 2,712 rich clusters north of about -27° declination, the original southern limit of the Palomar Observatory Sky Survey (POSS) made with the 48-inch Schmidt camera on Palomar Mountain. Which groups made Abell's cut? To qualify, a cluster needed at least 50 members within two magnitudes of the third-brightest galaxy. Furthermore, the cluster had to be sufficiently compact so these 50 members lay within a certain distance of the center (roughly 20 million light-years). Finally, nearby clus-

ters that were spread out over several POSS plates were excluded, which is why the Virgo Cluster has no Abell number.

Many large-telescope users are familiar with the rich northern clusters Abell 426 in Perseus, Abell 1367 in Leo, and Abell 1656 in Coma Berenices (which can be found at, respectively, right ascension $3^h 20^m$, decli-

Name	R.A. (2000.0)	Dec.	B mag.*	Size (')
IC 5349	$23^h 46^m 23^s$	$-28^\circ 00.2'$	15.3	0.9×0.2
IC 5350	$23^h 47^m 15^s$	$-27^\circ 57.5'$	14.8	0.7×0.6
IC 5353	$23^h 47^m 29^s$	$-28^\circ 06.6'$	14.0	1.4×1.0
IC 5354	$23^h 47^m 29^s$	$-28^\circ 08.2'$	14.8	0.8×0.4
PGC 72437	$23^h 47^m 43^s$	$-28^\circ 08.6'$	14.4	
PGC 72436	$23^h 47^m 43^s$	$-28^\circ 10.4'$	15.9	0.4×0.3
IC 5358	$23^h 47^m 44^s$	$-28^\circ 08.4'$	14.1	2.5×1.0
MCG-5-56-15	$23^h 47^m 56^s$	$-28^\circ 19.3'$	15.3	0.7×0.5
PGC 72466	$23^h 48^m 21^s$	$-28^\circ 13.9'$	15.0	
PGC 72469	$23^h 48^m 21^s$	$-28^\circ 07.0'$	15.5	
PGC 72471	$23^h 48^m 23^s$	$-28^\circ 04.5'$	14.7	

*Blue magnitudes from Green, Godwin, and Peach: *MNRAS* 243 (1990): 159.
Visual magnitudes are roughly 0.8 magnitude brighter.

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nation $+41.5^\circ$; $11^h 44^m$, $+20.0^\circ$; and $13^h 00^m$, $+28.0^\circ$). In my 17.5-inch scope under dark skies, these fields are breathtaking; scores of dim galaxies are visible within a few eyepiece fields of the central region.

But what about clusters south of Abell's cutoff?

To address this deficiency, Abell, Harold Corwin Jr., and Ronald Olowin examined Southern Sky Survey plates taken with the 48-inch Schmidt telescope at Siding Spring, Australia, using the same criteria as for the northern clusters. They identified more than 1,350 additional galaxy groups in a 1989 paper, finally providing complete sky coverage.

Northern observers have neglected these southern clusters — needlessly so. Yes, there are obstacles to observing faint galaxies low in the south. Your patience and observing skills may be tested by obstructions, poor seeing, and reduced transparency. But don't be deterred. Given a dark sky, the results can be surprising. Observing from latitude 38.5° north I've seen more than two dozen members of Abell 3526 in Centaurus, which hugs the horizon at -41° declination!

If you're ready to take the plunge, an excellent fall target is Abell 4038 in Sculptor. At -28° declination, it remains in undeserved obscurity.

This group was first noticed by Lewis Swift observing in 1896 with a 16-inch refractor from Echo Mountain in Southern California. He spotted four of the brighter members (IC 5350, IC 5353, IC 5354, and IC 5358). Herbert Howe re-observed the cluster a few years later with the 20-inch refractor at Chamberlin Observatory in Colorado; he recorded a fifth member, IC 5349. Studies in the 1970s revealed X-ray emission from hot gas in the cluster, a redshift ($z = 0.028$) yielding a distance of about 400 million light-years, and the presence of two supergiant, cD-type galaxies dominating the core. The cluster is roughly spherical and consists mostly of elliptical and S0-type galaxies. As with many other dense groups, it lacks gas-rich spirals like the Milky Way.

Ready to take a look? Although Abell 4038 may be poorly known, finding its location is a cinch; its core is $15'$ west of the 4.5-magnitude star Delta Sculptoris. My observations were made on two nights in November 1997 with a 17.5-inch f/4.4 Dobsonian at $220\times$ and $280\times$ in the Sierra Nevada foothills near Fiddletown, California. For a successful ob-

Abell 3744 in Capricornus is another southern cluster awaiting early-autumn observers.

Southern Abell Clusters for Northern Observers

Abell #	R. A. (2000)	Dec.	Const.	Brightest member
1060	10 ^h 36.6 ^m	-27° 31'	Hya	NGC 3309
3536	12 ^h 48.8 ^m	-41° 19'	Cen	NGC 4696
3565	13 ^h 36.6 ^m	-33° 58'	Cen	IC 4296
3574	13 ^h 49.1 ^m	-30° 18'	Cen	IC 4329
3656	20 ^h 00.8 ^m	-38° 34'	Sgr	IC 4931
3744	21 ^h 07.4 ^m	-25° 26'	Cap	NGC 7018
4038	23 ^h 47.7 ^m	-28° 08'	Scl	IC 5358

servation, you'll need good seeing and moderately high power to distinguish the small members from faint field stars.

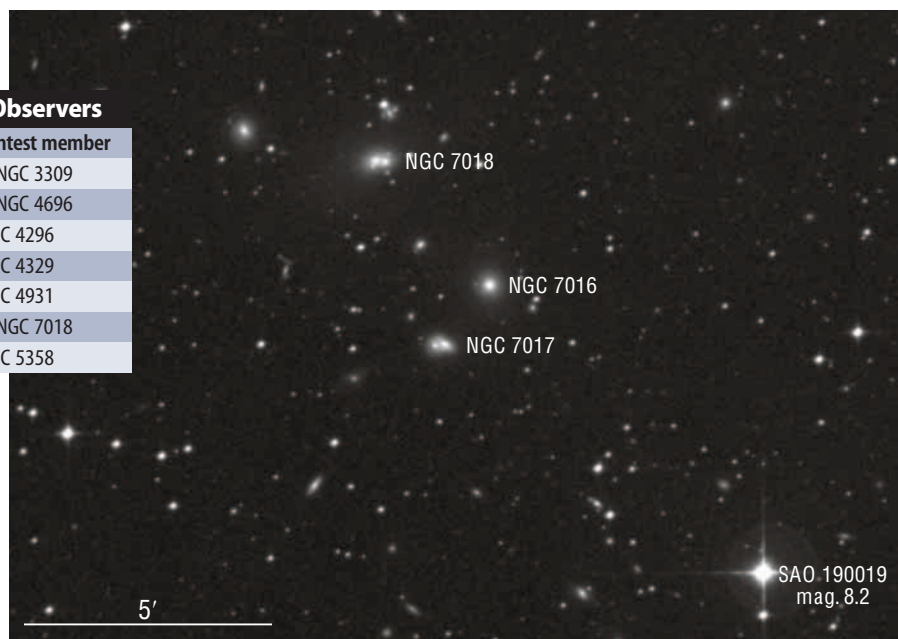
Sweeping west from Delta, I first picked up a pair of galaxies missed by both Swift and Howe. PGC 72471 is just 8' west-northwest of Delta, and once I moved the star out of the field it was immediately evident as a round, 20" glow just 1' south-southwest of a 13.5-magnitude star. Fainter PGC 72469, located 2.5' south, has a similar size but required averted vision to glimpse.

Next I dropped about 7' due south and found PGC 72466. This tiny elliptical was visible continuously with averted

vision at 280×. Drawing a line from Delta through PGC 72466 brought me to a dim S0 galaxy, MCG-5-56-15, less than 5' east of a 10th-magnitude star.

These were warmups for the dense central core of Abell 4038, which is dominated by the two giant cD galaxies IC

5358 and IC 5353. At first IC 5358 appeared as a single, irregularly shaped glow, but in moments of better seeing a double system was evident with two stellar nuclei just 20" apart. True to its reputation for having a voracious appetite, this bloated giant seemed to be feasting



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on an unfortunate companion (PGC 72437) embedded in the southwest side of its halo. With concentration I glimpsed another close companion, PGC 72436, a feeble 15" glow a mere 2' south.

The second-brightest member of the cluster is IC 5353, situated 4' northwest of IC 5358. With averted vision its irregular halo grows to nearly 1' across and appears weakly concentrated with a brighter center. The final cluster-core galaxy, IC 5354, is a challenging target 1.5' south of IC 5353. Large-scope users may spot a faint companion attached at the northeast edge.

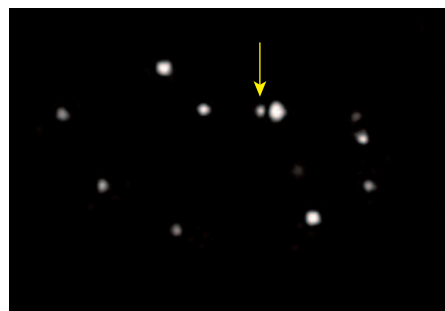
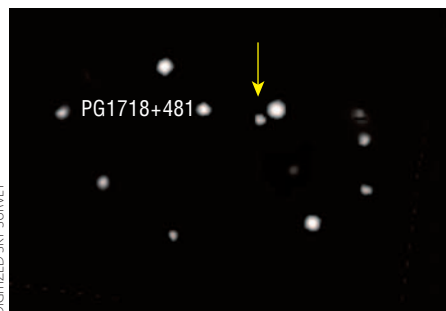
If you head 10' north-northwest from the tight trio in the core, you should run into IC 5350, which forms the vertex of an isosceles triangle with two 10th-magnitude stars 5' north and east-southeast. Although this galaxy is only 30" in diameter, a small, brighter core containing a stellar nucleus was evident.

On the western edge of the cluster is IC 5349, oriented roughly north-south and appearing twice as long as wide. Howe questioned whether this might be a nebulous double star, and indeed a nearly stellar companion is on the north end.

Looking for additional challenges in Abell 4038? There are several faint galaxies that I missed, including PGC 72393, 72451, 72464, 72473, and 72476. These should be visible in larger scopes or from locations with a better southern horizon. I'd be interested in hearing of your results.

Steve Gottlieb has been observing faint deep-sky objects for more than 20 years from dark sites in California. His last article, "Exploring Hickson Compact Groups," appeared in the March 1999 issue, page 110. He can be reached by e-mail at sgottlieb@telis.org.

■ observer's notebook | By Gary Seronik



A Serendipitous Discovery

Amateur discoveries are often the result of carefully planned search programs. Others, however, occasionally arise from simply being in the right place at the right time. The discovery made by Massachusetts amateurs Stephen LaFlamme, Dan Silva, and Ed Couture falls into the latter category. Last June, LaFlamme and Silva were hunting quasars from LaFlamme's Bridgewater backyard. They were able to spot two of these very faint objects in LaFlamme's 11-inch Celestron Schmidt-Cassegrain telescope (SCT). Recounts LaFlamme, "The following day, Dan mentioned our observations to Ed, who set up his 6-inch Astro-Physics refractor and CCD camera that evening to make an image of PG1718+481 in Hercules — one of the quasars we spotted the night before."

When he received Couture's shot the next day, LaFlamme used the Internet to download a comparison image (above left) from the Digitized Sky Survey. "As I studied the quasar and its surrounding region on the computer monitor, I noticed a faint star that changed its position by about 8 arcseconds in the 44 years between the images," he recalls. But what was the moving point of light? According to Brian Skiff of Lowell Observatory in Flagstaff, Arizona, the likeliest candidate is a star close

enough to the solar system to exhibit some proper motion. Such stars are not rare, but they are not usually discovered by chance.

Says LaFlamme, "We are still hoping that more professional astronomers will look into this so that we can learn more about this star. For now, it is exciting for us to know that this moving star has not been noted before!"

Omega North

In the August Observer's Notebook we told the tale of a pair of Canadians who managed to sight the grand globular Omega Centauri from Canadian soil at latitude 42° 32' north. We wrongly assumed this to be the most northerly sighting of the cluster. Massachusetts amateurs John Gall and Ralph Pass have seen Omega from a location even farther north — 42° 34' 32.2". They observed the cluster on the evening of May 24, 1998, from Magnolia, a seaside village with a view facing south over Massachusetts Bay. Using the "lifting" effect of atmospheric refraction to their advantage, Gall and Pass viewed the cluster with a 10-inch SCT and even managed to obtain a CCD image to prove the sighting. Remarks Pass, "We took six exposures of three minutes each. When displayed on the computer screen, the unmistakable signature of the globular cluster was seen."



Good-bye Mars, Hello Jupiter. On July 22nd, Maurizio Di Sciuillo imaged Mars when it exhibited a disk less than 10 arcseconds in diameter. In early May, when the planet was at its maximum size during the current apparition, Mars was 16.2 arcseconds across. Jupiter (right) was captured by Boston meteorologist and amateur astronomer Todd Gross in late June. Gross's image shows the Jovian moon Io and its shadow on the planet's disk.

